Exercise: Real Numbers

CS 255: Computer Architecture and Machine Level Programming

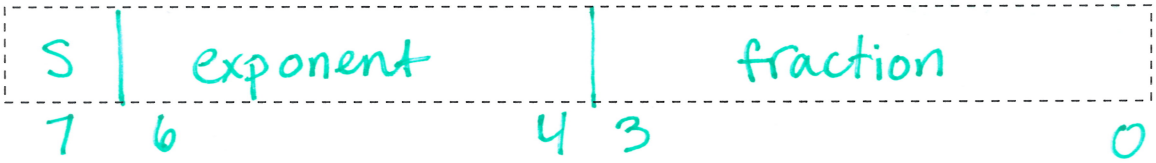
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In this exercise, your task is to design your own floating point standard using 8 bits.

1. Draw a bit diagram of your standard.



   (a) How many bits represent the sign?

   1

   (b) How many bits represent the exponent?

   3

   (c) How many bits represent the mantissa?

   4

2. Choose an **excess/bias** to encode the exponent? (i.e. IEEE 764 uses excess-127)

   excess-3

   keep 000 and 111 as special cases, so unsigned exponent stores values 1-6. Excess-3 encoding makes the exponent range -2 to 3

3. What kinds of numbers does your format represent well?

it represents numbers between 0.25 and 15.5, both positive and negative

it represents fine differences between small values

4. What kinds of numbers does your format fail to represent?

no numbers outside the range [0.25, 15.5]

it does not represent fine differences between large values

5. How do your design choices contribute to what kinds of numbers your format can and cannot represent?

more exponent bits: more precision, less range

more fraction bits: more range, less precision

sign bit: represent positive numbers or not

bias shifts range:

- smaller bias: larger magnitudes
- larger bias: smaller magnitudes